MILITARY SPECIFICATION

# SAW, M249 BLANK AMMUNITION MAGAZINE ASSEMBLY AND FEEDSTRAP

This specification is approved for use by the US Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification contains requirements and quality assurance provisions for the SAW, M249 Blank Ammunition Magazine Assembly and Feedstrap. The assembly consists of a magazine with a cover; the feedstrap is not a part of the assembly.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this document to the extent specified herein.

SPECIFICATIONS

MILITARY

| | | |
|---|---|---|
| MIL-A-2550 | - | Ammunition, General Specification for |
| MIL-A-48078 | - | Ammunition, Standard Quality Provisions General Specification for |
| MIL-C-60616 | - | Cartridge, Blank, 5.5mm, M200 |
| MIL-W-63150 | - | Weapons and Support Materiel Standard Quality Assurance Provisions for |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-L-63460 - Lubricant, Cleaner and Preservative for Weapons and Weapon Systems

STANDARD

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein.

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER (ARDC)

| | | |
|---|---|---|
| 9357903 | - | SAW, M249 Blank Ammunition Magazine Assembly |
| 9352576 | - | Cover |
| 9357902 | - | Magazine |
| 9352578 | - | Feedstrap |

(Copies of specifications, standards, handbooks, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Other publication. The following document(s) form a part of this specification to the extent specified herein.

ASTM D 3951-82 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103)

2.1.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 Magazine assembly and feedstrap. The magazine assembly and feedstrap shall comply with all requirements of drawing 9357903, 9352578 all associated drawings, and with all requirements specified in applicable specifications and standards.

3.2 Materials. All materials shall be in accordance with the applicable drawings and specifications. Certificates of conformance shall be included with all inspection quantities.



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Information Handling Services

3.3 First article. When specified in the contract or this document, a sample shall be subjected to first article inspection.

3.4 Functioning. The magazine shall be capable of being loaded with government accepted ammunition linked with government accepted M27 links. The magazine cover and feedstrap shall be attached. The loaded assembly shall be attached to a M249 machine gun and functioned without any malfunctions attributable to the assembly. Prior to firing the last 50 cartridges, the weapon feed cover shall be released and the free length of cartridges allowed, if possible, to fall back into the magazine, should this occur, it will be considered a malfunction. After functioning, the magazine shall be capable of being detached from the weapon without difficulty.

3.5 Dovetail mount. The magazine assembly (inert loaded) shall remain attached to the weapon mount fixture following the drop of a 5 pound weight, from a height of 12 inches, onto the magazine at a distance of 5 inches from the dovetail.

3.6 Drop test. When an inert loaded magazine assembly is dropped from a height of 12 inches, the assembly shall not deform to the extent that its functional performance is impaired. This includes any separation of the cover from the magazine which results spillage of the ammunition, or cracking of the magazine that would prevent its proper functioning.

3.7 Spacer. The spacer shall not dislodge from its seated position during the drop test or any other testing.

3.8 Locking tab. The locking tab shall have sufficient strength and spring action to assure that the magazine will remain attached to the weapon mount fixture following the drop of a 2 pound weight onto the impact device from a height of 12 inches. When depressed, the locking tab shall completely disengage from the weapon mount so as not to hinder removal of the magazine assembly from the weapon mount.

3.9 Magazine component interface. The following pairs of items shall assemble and disassemble without the use of tools and without damage to either components.

a. Magazine and cover

b. Magazine and feedstrap

c. Magazine and weapon mount



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3.10 Temperature requirements. All requirements stated for the magazine assembly shall be met under ambient conditions and also after the assembly has been temperature conditioned for at least two hours and tested at:

a. 125° - 5°F

b. -65° + 5°F

3.11 Workmanship. Workmanship shall be in accordance with good manufacturing practice and shall in no way impair the functional performance of the magazine. Magazines shall be free of cracks, extraneous material, and irregularities in the surfaces. Metal parts shall not be rusted. The drain holes shall not be obstructed. The cartridge silhouette shall be in the proper location and orientation. The area about the flap and dovetail shall be uniform in structure and free of sink marks greater than .015 in depth. The gate shall be flush not to exceed .050 in height or below the surface. The raised matted surfaces on the cover shall be uniform in size and surface finish. In addition, MIL-W-63150, (Workmanship) shall apply.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsbility for inspection. In accordance with MIL-W-63150 (Responsibility for inspection).

4.1.1 Inspection equipment. In accordance with MIL-A-48078 (Inspection equipment) and MIL-A-2550 (Test and measuring equipment).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First Article Inspection. A First Article shall be required when any of the following occurs:

a. A First Article is required in accordance with the contract requirements.

b. A new or reworked mold is to be used for the first time.

c. The batch of material is changed to a batch of material that has not been used previously.



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4.3.1 Submission. The contractor shall submit a first article sample in quantities as indicated below for each mold cavity to be used in production, in accordance with the First Article requirements of the contract.

| Item | Quantity | Drawing # |
|---|---|---|
| Magazine | 50 | 9357902 |
| Cover | 50 | 9352576 |
| Feedstrap | 50 | 9352578 |

4.3.2 Inspections to be performed. The First Article sample shall be subjected to all inspections listed in Table 1, First Article Inspection. In addition, first article assemblies, components, and test specimens may be subjected by the Government to any or all of the examinations and tests specified in this specification and to any or all requirements of the drawings.

4.3.3 Rejection. Failure of any item to meet the drawing requirements, acceptance criteria of Table 1, or the referenced paragraphs shall result in rejection of the First Article. In addition MIL-A-48078 (Rejection) shall apply.

4.4 Quality conformance inspection.

4.4.1 Lot formation. Lot formation shall be in accordance with MIL-A-48078 (Inspection Lot Formation).

4.4.2 Lot submission. The product shall be submitted in accordance with MIL-STD-105.

4.4.3 Inspections to be performed. Each lot shall be subjected to all inspections listed in 4.6.2. The government reserves the right to waive high and low temperature testing following the acceptance of three successive production lots.

4.4.4 Rejection. Failure of any sample to meet the acceptance criteria of para 4.6.2 and referenced paragraphs shall result in rejection of the lot.

4.5 Certificates of conformance (COC). COC's representing the contractor's own criteria and inspection results for control of the following items shall be made available before first article or lot acceptance will be accepted.

Item

a. Material

b. Pin (for magazine)



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c. Spacer (for magazine)

The COC's shall be present in the inspecton reports.

4.6 Dimensional settlement of plastic is required before inspection. All inspections for government acceptance shall be performed at least 48 hours after manufacture of the item.

4.6.1 Examination for defects (major and minor). Examination for major and minor defects shall be performed on an individual basis as specified, in accordance with Table II using applicable sampling plans and acceptance criteria of MIL-STD-105. The Acceptance Quality Level (AQL) shall be as specified in 4.6.2. All non-conforming cartridges (or components) shall be rejected.



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TABLE 1. First article inspection

**CLASSIFICATION OF DEFECTS & TESTS** MIL-C-70471(AR)

| PARAGRAPH | TITLE: Magazine assembly with Feedstrap | | SHEET 1 OF 1 | | DRAWING NUMBER: 9357903 |
|---|---|---|---|---|---|
| | | | | | NEXT HIGHER ASSEMBLY: N/A |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| | Workmanship/Visual examinations | 1/ | 2/ | 3.11 | 4.7.7 |
| | Magazine component interface | 1/ | 2/ | 3.9 | 4.7.1 |
| | Locking tab | 1/ | 2/ | 3.8 | 4.7.2 3/ |
| | Dovetail mount | 1/ | 2/ | 3.2 | 4.7.3 3/ |
| | Drop test | 1/ | 2-3 | 3.6 | 4.7.4 3/ |
| | Functioning | 4/ | 2/ | 3.4 | 4.7.5 3/ |
| | Temperature requirments | 1/ | 2/ | 3.10 | 4.7.6 |
| | DIMENSIONAL SURVEY | | | | |
| | Magazine Assembly a/ | 5 | 0-1 | 3.1 | 9357903 |
| | Feedstrap | 5 | 0-1 | 3.1 | 9352578 |

NOTES: 1/ A total of 30 assemblies are tested; 10 at each temperature condition.
2/ No failures are permitted.
3/ Loading of magazines shall be as specified in paragraph 4.7.8.
4/ A total of 15 assemblies are tested; 5 at each temperature condition.
a/ Magazine assembled to a cover

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## QUALITY CONFORMANCE INSPECTION

## CLASSIFICATION OF DEFECTS & TESTS

MIL-C-70471 (AR)

| PARAGRAPH | TITLE | SHEET 1 OF 1 | DRAWING NUMBER |
|---|---|---|---|
| 4.6.2 | Magazine Assembly with Feedstrap | | 9357903 |
| | | | NEXT HIGHER ASSEMBLY N/A |

| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
|---|---|---|---|---|---|
| | AMBIENT TESTING | | | | |
| | Workmanship/Visual examinations | 3/ | 4.0 | 3.11 | 4.7.7 |
| | Magazine component interface | 3/ | 1.5 | 3.9 | 4.7.1 |
| | Locking tab | 3/ | 1.5 | 3.8 | 4.7.2 2/ |
| | Dovetail mount | 3/ | 1.5 | 3.5 | 4.7.3 2/ |
| | Drop test | | 0-1 | 3.6 | 4.7.4 2/ |
| | SPECIAL TESTS | | | | |
| | Drop test (high temperature) | 4/ | 2.5 | 3.6, 3.10 | 4.7.4 2/ |
| | Drop test (low temperature) | 4/ | 2.5 | 3.6, 3.10 | 4.7.4 2/ |
| | Functioning | 1/ | 0-1 | 3.4 | 4.7.5 2/ |
| | DIMENSIONAL SURVEY | | | | |
| | Magazine assembly | 3 | 0-1 | 3.1 | 9357903 |
| | Feedstrap | 3 | 0-1 | 3.1 | 9352578 |

NOTES: 1/ Three assemblies shall be tested at ambient, three assemblies shall be tested under high temperature conditions and another three assemblies shall be tested under low temperature conditions.
2/ Loading of magazines shall be as specified in paragraph 4.7.8.
3/ Determined by MIL-STD-105, inspection level S-3, single sampling plan.
4/ Determined by ML-STD-105, inspection level S-3, double sampling plan.

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4.7 Method of inspection.

4.7.1 Magazine component interface. As stated in the requirement of Section 3, magazines shall be assembled to and disassembled from the cover, feedstrap, and weapon mount without the use of tools and without damage. Conformance to these requirements may be examined for during Locking Tab, and Dovetail Mount testing.

4.7.2 Locking tab. A linked belt of 200 M200 inert blank cartridges shall be loaded into the magazine and the cover attached. The assembly shall be attached to the test fixture and the locking tab shall be examined for proper assembleability. The container shall be disassembled from the test fixture and while doing so, the locking tab shall be forced downward to the maximum extent of its travel. The dovetail and locking tab shall then be lubricated generously with "Break Free CLP" lubricant, MIL-L-63460, and attached to the test fixture again. The impact device shall be seated against the dovetail. A two pound weight shall be dropped onto the impact device from a height of 12 inches. The magazine assembly must remain attached to the test fixture, the locking tab shall remain functionable (assemblable and disassemblable) and, shall not be cracked in order for the test result to be acceptable.

4.7.3 Dovetail mount. A linked belt of 200 M200 inert blank cartridges shall be loaded into a magazine and the cover attached. The dovetail shall be lubricated generously with "Break Free CLP" lubricant, MIL-L-63460, and the magazine attached to the test fixture. The dovetail shall be examined for proper fit (interface requirement). A five pound weight shall be dropped from a height of 12 inches onto the the magazine five inches away from the dovetail (directly opposite from the dovetail). The magazine must remain attached to the fixture for the test result to be acceptable.

4.7.4 Drop test. A linked belt of 200 M200 inert blank cartridges with feedstrap attached shall be loaded into a magazine and the cover attached. The assembly shall be dropped onto solid concrete three times. Once on the cover edge designated as 3-4, once on the cover corner designated as 2, and once on the corner opposite the cover designated as 7. See drawing 9357903 for designation of these drop locations. Drop height shall be 12 inches in all cases. The assembly must comply with the stated requirement for the test results to be acceptable.



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4.7.5 Function firing. Each magazine shall be loaded with a linked belt of 200 government approved M200 blank confroming to MIL-C-60616 with government approved M27 links and a feedstrap attached. The cover shall be assembled to the magazine and the assembly shall be temperature condition as required in Section 3. Each magazine shall be fired using a M249 machine gun in burst of 5-7 rounds at a rate of 85 shots per minute. The weapon barrel shall be cooled to ambient following the firing of each 200 round group.

4.7.6 Temperature extremes. Test and examinations shall be performed in the same manner as for ambient conditions with all of the following modifications.

a. All magazines, covers, feedstraps and linked cartridges shall be temperature conditioned as an assembly at the specified temperature for a minimum of two hours before the start of testing.

b. Testing shall be performed immediately upon removal of the conditioned items from the conditioning box. A delay of more than two minutes is unacceptable and shall require reconditioning for 30 minutes. Immediately after each test or examination, the item shall be returned to the conditioning box so that it will remain temperature conditioned for the next test to be performed on it.

c. After the initial two hours of temperature conditioning, items need only be reconditioned for 30 minutes between each test provided the conditioned item is not exposed to ambient temperatures for more than five minutes during the test. Otherwise two hours of conditioning is required.

4.7.7 Workmanship/Visual examinations. The magazine assemblies and components shall be carefully examined for any defects that could affect the serviceability of the item.

4.7.8 Loading of magazines. With the open end of the magazine facing up, the single-loop end of the linked belt is placed inside near the image at the bottom of the magazine, with the cartridge tips facing the same direction as indicated by the image. The linked belt is then loaded into the magazine in a zig zag fashion taking care that all rows are level to the bottom surface. Do not leave any voids at the end of each row. When the entire belt is loaded into the magazine, the end of the belt is brought to the flap opening so that it can be attached to the feedstrap. The feedstrap is inserted into the magazine through the flap opening so that the double loops of the link match to the partial cylinder of the feedstrap. The double loops of the link are then slipped onto the partial cylinder and rotated to snap into place. The cover is then snapped onto the magazine.



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5. PACKAGING

5.1 Preparation for delivery. Packaging and packing shall be commercial as specified in Standard Practice for Commercial Packaging ASTM D 3951-82.

6. NOTES

6.1 Intended use. Assemblies and components covered by this document are intended for use with the M249, 5.56mm machine gun and associated 5.56mm ammunitions.

6.2 Ordering data. See MIL-A-48078 (Ordering data).

Custodian:
Army-AR

Preparing Activity:
Army-AR

(Project 1005-A649)



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